SECTION 1 — REVISION NOTES
GCSE AQA PHYSICS: ELECTRICITY

(Bullet-point, mark-scheme terminology, Word-ready)

ELECTRIC CURRENT

- **Current = rate of flow of charge**
- Charge measured in **coulombs (C)**
- Current measured in **amperes (A)**

Formula

$I = Q / t$

Where:

- I = current (A)
- Q = charge (C)
- t = time (s)

POTENTIAL DIFFERENCE (VOLTAGE)

- Potential difference = **energy transferred per unit charge**
- Measured in **volts (V)**

Formula

$V = E / Q$

Where:

- V = potential difference (V)
- E = energy transferred (J)
- Q = charge (C)

RESISTANCE

Resistance = opposition to current flow.

Measured in **ohms (Ω)**

Formula

$R = V / I$

OHM'S LAW

For an ohmic conductor at constant temperature:

Current is **directly proportional** to potential difference.

Graph:

- Straight line through origin on V–I graph

SERIES CIRCUITS

- Same current everywhere
- Potential difference shared
- Total resistance = sum of resistances

If one component fails → circuit breaks.

PARALLEL CIRCUITS

- Same potential difference across each branch
- Current splits between branches
- Total resistance decreases when branches added

Used in household wiring.

RESISTANCE IN SERIES

R_total = R1 + R2 + R3

FILAMENT LAMP BEHAVIOUR

- Resistance increases as temperature rises
- V–I graph curves

Exam phrase:

✔ resistance increases due to heating

THERMISTOR

- Temperature increases → resistance decreases

Used in:

- temperature sensors

LDR (LIGHT DEPENDENT RESISTOR)

- Light intensity increases → resistance decreases

Used in:

- automatic street lights

MAINS ELECTRICITY

UK mains:

- 230 V AC
- frequency 50 Hz

Alternating current:

- direction constantly changes

Direct current:

- flows in one direction only

LIVE / NEUTRAL / EARTH WIRES

Live wire

- carries alternating potential difference

Neutral wire

- completes the circuit

Earth wire

- safety wire
- carries current only if fault occurs

PLUG SAFETY

Fuse:

- melts if current too large
- breaks circuit

Plastic casing:

- electrical insulator

Cable grip:

- prevents wires being pulled loose

ELECTRICAL POWER

Power = rate of energy transfer.

Formula

$P = V \times I$

Also:

$P = E / t$

Unit = watt (W)

ENERGY TRANSFERRED BY APPLIANCE

Formula

$E = P \times t$

If using V and I:

$E = VIt$

Energy measured in **joules (J)**

DOMESTIC ELECTRICITY COST

Suppliers charge per **kilowatt-hour (kWh)**

1 kWh = energy used by 1000 W appliance in 1 hour.

REQUIRED PRACTICAL — CURRENT–VOLTAGE

Typical method:

- connect resistor to power supply
- measure current with ammeter
- measure voltage with voltmeter
- change voltage
- record readings
- plot graph

COMMON GCSE ELECTRICITY EXAM MISTAKES

Students lose marks for:

- mixing up current and voltage
- forgetting units
- drawing voltmeter in series (WRONG)
- drawing ammeter in parallel (WRONG)
- saying current used up
- not converting time to seconds
- forgetting kW to W conversion

EXAMINER KEYWORDS THAT SCORE MARKS

Use:

- “rate of flow of charge”
- “energy transferred per unit charge”
- “current splits in parallel circuit”
- “potential difference shared in series”
- “resistance increases due to heating”